

Edge Deployment Planning API

A CAMARA proposal for pre-deployment rollout planning

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1. The Gap

CAMARA already covers key edge-cloud steps:

- **OED**: runtime edge target selection
- **EAM**: deployment and lifecycle management
- **AED**: runtime endpoint discovery

Missing step: pre-deployment planning.

Open question:

Before deployment starts, which edge options are feasible in a target area?

2. Example Use Case - Multi-region cloud gaming rollout

A provider wants to assess where a **public** latency-sensitive gaming service can be launched.

The Problem: without a common planning API, the developer must discover and compare edge options operator by operator.

Expected API output

- feasible edge zones in the target area
- which operators / providers can serve them
- estimated latency envelope

3. Why This Is Not Covered By Optimal Edge Discovery (OED)

OED

- runtime / near-runtime decision
- active device or session context
- returns the **best** target edge

Edge Deployment Planning API

- pre-deployment decision
- no active device required
- returns the **feasible set**

In one line: OED is optimization; Edge Deployment Planning is viability.

4. What the API Should Do

Given:

- a **target area**
- an **application profile**
- a set of **constraints**

Return:

- feasible **edge zones**
- relevant **edge provider/ mobile network operator information**
- estimated **performance metrics**

5. Position in the Edge Cloud Family

API	Purpose
OED	best target selection
EAM	deployment / lifecycle
AED	runtime endpoints
Edge Deployment Planning API	pre-deployment feasibility

This fills a new lifecycle step rather than duplicating an existing one.

6. Design Principle

Developer-facing abstraction

The API should expose:

- candidate edge options
- viability results
- estimated planning metrics

The API should **not** expose:

- DNN
- S-NSSAI
- internal routing policies
- operator topology

7. Why It Matters

- helps developers decide **whether to launch**
- supports **rollout planning** across areas
- enables **multi-operator aggregation**

Goal: allow developers and platforms to make informed go / no-go rollout decisions before committing engineering work.

Thank you

Discussion points

- Is pre-deployment feasibility a useful missing capability?
- Is the right path a new sibling API, rather than an OED extension?